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LAB-TASK-5

Task 5 – OpenCV Image Processing and Computer Vision

Introduction

OpenCV (Open Source Computer Vision Library) is a powerful library used for image processing, computer vision, and real-time applications. In this task, different OpenCV techniques were implemented in Python to perform image enhancement, filtering, noise removal, and color detection using webcam input.

The implementations were based on the OpenCV Python tutorial from GeeksforGeeks.

Technologies Used

- Python
- OpenCV
- NumPy
- Matplotlib

Steps Implemented

1. Image Processing and Enhancement

The image was first loaded using OpenCV and displayed using Matplotlib. Converting the image to grayscale allowed easier processing and visualization of pixel intensity.

This step helps in preparing the image for further processing operations.

2. Image Segmentation and Thresholding

Image segmentation techniques were studied to separate objects from the background. These techniques help divide an image into meaningful regions for further analysis.
(The last two methods from this section were not implemented.)

3. Morphological Operations and Filtering

Morphological operations were applied to modify the structure of objects in the image.

Two operations were used:

Erosion
This operation shrinks bright regions of the image and removes small noise.

Dilation
This operation expands bright regions and helps restore object shapes after erosion.

These operations are useful for cleaning and refining images.

4. Advanced Image Manipulation and Background Subtraction

Advanced filtering techniques were applied to improve image quality.

Bilateral Filtering
A bilateral filter was used to smooth the image while preserving edges.

Noise Removal
The Fast Non-Local Means Denoising method was used to remove noise from the image while keeping important details intact.

These techniques improve image clarity before performing further analysis.

5. Feature Detection and Description

Feature detection techniques were studied to identify important structures and patterns in images such as edges, shapes, and keypoints. These features are useful in object recognition, tracking, and image matching.

6. Feature Detection and Analysis

Feature analysis techniques help analyze detected features and extract useful information from images. These techniques are widely used in computer vision tasks such as object detection and pattern recognition.

7. Real-Time Color Detection using Webcam

A webcam feed was captured using OpenCV. The video frames were converted from BGR color space to HSV color space. A color mask was created to detect blue objects in the frame.

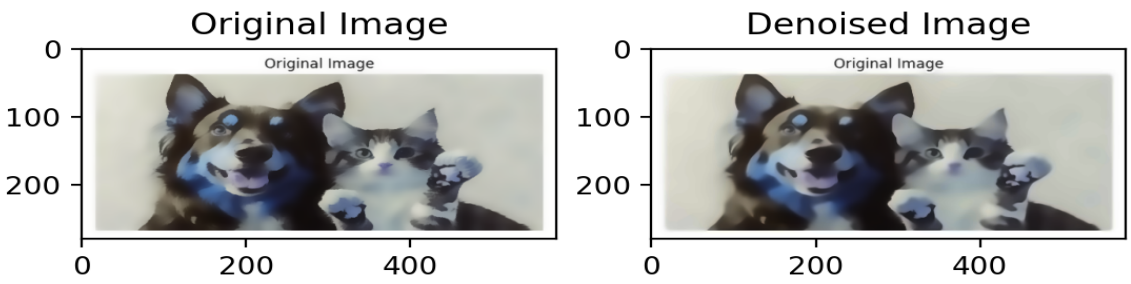
The mask was applied to the original frame to highlight only the blue regions while removing other colors. The processed frames were displayed in real time.

The program stops when the user presses the **q** key.

Example run

Original Image
Original Image

After Erosion
Original Image


Original Image
Original Image
0
100
200
0
200
400
Denoised Image
Original Image
0
100
200
0
200
400